CHAPTER

11

Fourier Transform Pairs

For every *time domain* waveform there is a corresponding *frequency domain* waveform, and vice versa. For example, a rectangular pulse in the time domain coincides with a sinc function [i.e., sin(x)/x] in the frequency domain. Duality provides that the reverse is also true; a rectangular pulse in the frequency domain matches a sinc function in the time domain. Waveforms that correspond to each other in this manner are called *Fourier transform pairs*. Several common pairs are presented in this chapter.

Delta Function Pairs

For discrete signals, the delta function is a simple waveform, and has an equally simple Fourier transform pair. Figure 11-1a shows a delta function in the time domain, with its frequency spectrum in (b) and (c). The magnitude is a constant value, while the phase is entirely zero. As discussed in the last chapter, this can be understood by using the expansion/compression property. When the time domain is compressed until it becomes an impulse, the frequency domain is expanded until it becomes a constant value.

In (d) and (g), the time domain waveform is shifted four and eight samples to the right, respectively. As expected from the properties in the last chapter, shifting the time domain waveform does not affect the magnitude, but adds a linear component to the phase. The phase signals in this figure have not been *unwrapped*, and thus extend only from $-\pi$ to π. Also notice that the horizontal axes in the frequency domain run from -0.5 to 0.5. That is, they show the *negative* frequencies in the spectrum, as well as the *positive* ones. The negative frequencies are redundant information, but they are often included in DSP graphs and you should become accustomed to seeing them.

Figure 11-2 presents the same information as Fig. 11-1, but with the frequency domain in *rectangular form*. There are two lessons to be learned here. First, compare the polar and rectangular representations of the